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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – APRIL 2023

Program	:	**B.E :- Computer Science and Engineering**	**Semester**	:	**III**
Course Name	:	**Data Structures**	**Max. Marks**	:	**100**
Course Code	:	**CS33**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Write a C function to dynamically allocate two dimensional array. Mention how the pointer to pointer concept is used there.	CO1	(07)
	b)	Explain sparse matrices and write function to transpose sparse matrix with an example.	CO1	(07)
	c)	Analyze any five string related library functions with appropriate examples.	CO1	(06)
2.	a)	Identify the criterias to be satisfied by algorithms.	CO1	(07)
	b)	Illustrate KMP pattern matching algorithm with an example.	CO1	(07)
	c)	Write a C function to perform the fast transpose operation over a sparse matrix.	CO1	(06)

UNIT - II

3.	a)	Apply infix to postfix conversion algorithm to convert the following infix expressions to postfix expressions: i. (A+B)*C(D/(J+D)) ii. A+ (B*C-(D/E^F)*G)*H	CO2	(08)
	b)	Why do we need dynamic arrays for Stack implementation? Justify your answer with an example.	CO2	(06)
	c)	Illustrate add and delete operations on a Circular queue with suitable examples.	CO2	(06)
4.	a)	Use the postfix evaluation algorithm to evaluate the following expression: i. 5 6 2 + * 12 4 1 – ii. 2 3 * 5 4 * + 9 - Tabulate the input, stack contents and output.	CO2	(08)
	b)	Identify the limitations of a linear Queue with suitable examples.	CO2	(06)
	c)	Write the C function to implement multiple stacks push() and pop() operations.	CO2	(06)

UNIT - III

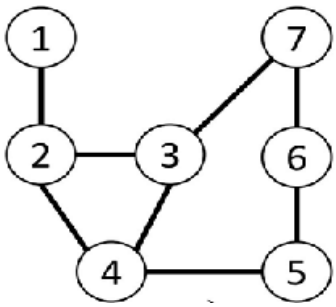
5.	a)	Write a graphical representation to show the basic operations of a List.	CO3	(06)
	b)	Describe the advantages of linked representation of data over sequential representation with suitable examples.	CO3	(06)
	c)	Write a C functions to implement Stacks using a singly linked list with header nodes.	CO3	(08)
6.	a)	Show the sparse matrix representation using linked list.	CO3	(06)
	b)	Illustrate a doubly linked circular list with a header node using a neat diagram.	CO3	(06)
	c)	Write C functions to implement multiple Queues represented as a linked list.	CO3	(08)

UNIT- IV

7.	a)	Define binary tree. Write any three properties of binary trees.	CO4	(06)
	b)	Differentiate between max heap and min heap with an example.	CO4	(06)
	c)	Write an iterative C function to search for an element in a binary search tree.	CO4	(08)
8.	a)	Illustrate with examples how to transform a forest into a binary tree.	CO4	(06)
	b)	Explain the memory representation of a threaded binary tree with an example.	CO4	(06)
	c)	Construct a binary search tree stepwise for the following list of numbers: 24, 12, 6, 18, 30, 54, 7, 10, 42, 36. Perform inorder, preorder and postorder traversal on the constructed binary search tree.	CO4	(08)

UNIT - V

9.	a)	Inspect the two ways of representing the following graph.	CO5	(07)



	b)	List out the operations over the following and describe: i) Single ended priority queue ii) Double ended priority queue.	CO5	(07)
	c)	Describe about weight biased leftist tree with appropriate example.	CO5	(06)
10.	a)	Write a c function to perform depth first search over a graph. Show the dfs() function using an example.	CO5	(07)
	b)	Illustrate height biased leftist tree with an example.	CO5	(07)
	c)	Inspect the balance factor usage in AVL tree with an example.	CO5	(06)

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